

Prateek Tiwari

AC 6, Level 1, Wing B, ISB Hyderabad, India - 500111

E-Mail | [LinkedIn](#) | [GitHub](#) | (+91)8840627870

RESEARCH PROFILE

I am a Research Associate (Pre-Doctoral) at the Indian School of Business (ISB), working with Prof. Deepa Mani on projects that sit at the intersection of technology and work. I build large-scale data infrastructures and analytic tools to understand how AI and digital platforms are reshaping jobs, skills, and how organisations learn. My work combines NLP, LLMs and embedding methods to model how skill bundles form, and change over time. More broadly, I am interested in human AI interaction and information systems, and aim to ground this work in empirical evidence that informs organisations and policy makers to adapt to these changes.

EDUCATION

- | | |
|---|----------------------|
| International Institute of Information Technology (IIIT-B) | Bangalore, India |
| <i>PG Diploma in Data Science (Deep Learning Specialization), GPA: 3.31/4 (WES: 3.61/4)</i> | 2021 - 2022 |
| Indian Institute of Petroleum and Energy (IIPE) | Visakhapatnam, India |
| <i>Bachelor of Technology in Petroleum Engineering, GPA: 8.33/10 (WES: 3.88/4)</i> | 2016 - 2020 |
| – Graduated among the top 10 students in the department. | |
| – Admitted through the JEE Advanced (2016), ranking in top 7% of the JEE Mains qualified candidates nationwide. | |

RESEARCH & PROFESSIONAL EXPERIENCE

- | | |
|---|---------------------------|
| Indian School of Business (ISB) | Hyderabad, India |
| <i>Research Associate (Pre-Doctoral)</i> | <i>Dec 2023 – Present</i> |

Primary Research Projects

- Worked on creating a high frequency labour market database for India, by developing an AI-ML pipeline to extract, standardize, and classify job postings across geographies, enabling scalable and granular labour market analytics.
- Developed a skill ontology and knowledge network from the jobs by integrating structured LLM-based extraction engine with Neo4j graph and created algorithms to align taxonomies from O*NET, NCO, NIC, and Lightcast.
- Validated and refined embedding based mapping and entity disambiguation techniques to improve skills and tasks standardization, enhancing coverage and precision for downstream analysis.
- Assisted in integrating unit level survey data (PLFS) with the job postings database to assess representativeness across occupations, industries and geographies. Also constructed wage distributions to perform robustness checks and evaluate alignment between demand and supply side labour market indicators.
- Synthesized research findings into scholarly and policy outputs with [Prof. Deepa Mani](#) (ISB) and created initial drafts for government grant proposals.

Additional Research Projects

- Assisted early stage data sourcing and exploratory analysis, estimating gender bias in Academia with [Prof. Vasundhara](#) (ISB) and [Prof. Aditya](#) (UT-Dallas), contributing to study design and visualization.
- Investigated geographic and geopolitical representation biases in LLMs using frameworks such as *Dataset Geography*, *GPPF*, and *GeoMLAMA* and evaluated the feasibility of IP-based geolocation for spatial bias detection.

Contributions to School Interventions

- Worked closely with ISB Career Advancement Services to translate research findings into curricular interventions for the PGP and PGP-YL programs by mapping skill demand across roles, industries, and cohorts.
- Standardized, integrated, and validated institutional databases across admissions and program-analytics frameworks, improving data reliability and operational efficiency for institutional reporting and program design.

- | | |
|--|--|
| Tata Consultancy Services (TCS) | Riyadh, Saudi Arabia (Onsite Deputation) |
| <i>System Engineer</i> | <i>Jul 2021 – Dec 2023</i> |

- Contributed to a large-scale digital-transformation program for a leading Middle-East telecom client; recognised with a Performance Excellence Award (2023).
- Applied ML and NLP pipelines to CRM datasets to extract product and customer insights, which directly helped in informed product development decisions.
- Automated data workflows using VBA and SQL, reducing processing time and improving accuracy. These measures contributed to a 44% reduction in time-to-market and a 20% increase in implemented change requests.
- Worked in a fast track Oracle CRM team, which was awarded "Best Team" for two consecutive years (2022, 2023).

WORKING PAPERS

The Indian Skills Database and Taxonomy

with Prof. Deepa Mani and Dr. Nishant Chadha

Draft available upon request.

- Constructed a national scale, labour market dataset and accompanying taxonomy framework using LLM and NLP based extraction and aligned occupations and skills across NCO, NIC, and Lightcast classifications. The core methodological approach was presented at the *Conference on the Digital Economy (CODE)*, 2025.

Skill Value Prediction Using Job Postings Data

with Prof. Deepa Mani and Munipalle Sai Nikhila

([code](#))

- Assisted in creating a framework to infer the market value of individual and compositional job skills using regression and DL architectures, including composition aware skill embeddings and graph based SSCN models. Also extended the model to the full national dataset to estimate dynamic skill premiums across occupations.

BOOK CHAPTER

Sports Matter

with Prof. Deepa Mani, *Rules of the game (In Progress)*

- Contributed analytical inputs and data insights for the chapter's empirical sections, supporting the development of the final narrative.

RELEVANT COURSEWORK

- **PhD (FPM) Credited Courses (ISB)**: Econometrics A (Grade: 3.5/4), Econometrics B (Grade: 4/4)
- **Graduate (IIIT-B)**: Data Toolkit, Machine Learning I & II, Deep Learning (Specialization Course)
- **Undergraduate (IIPE)**: Probability & Statistics, Numerical Methods, Design of Experiments, Data Analytics & AI for Process Industry, Economics, Project Engineering and Management

CONFERENCE PRESENTATION

- **Tiwari, P.**, Mani, D., Chadha, N., & Garimella, P. (2025, Jan). *Bridging Skill (Data) Gaps: Methodology and Results for an Indian Software Skills Database*. Presented at CODE (Varanasi). ([slides](#)) ([paper](#))

ACADEMIC ENGAGEMENT & SERVICE

- ISB Data Science Summit, Hyderabad, Nov 2024 & Oct 2025.
- Analytics Convergence Workshop (JP Morgan Chase & IIDS), ISB Hyderabad, Oct 2025.
- Production and Operations Management (POMS) Workshop, ISB Hyderabad, Jul 2025.
- Data Users Conference (NSO, MoSPI & ISB), Hyderabad, May 2025.
- Volunteer, *Workshop on Information Systems and Economics (WISE)*, Hyderabad, Dec 2023.
- Volunteer, *Workshop on Information Technologies and Systems (WITS)*, Hyderabad, Dec 2023.

ACADEMIC PROJECTS

Caption Generation Model using Flickr8K Dataset

2021

International Institute of Information Technology, Bangalore

([code](#))

- Developed an encoder-decoder attention model to generate image captions for visually impaired users, evaluated via BLEU score (0.67).

Impact of Crises on Global Oil Prices

2020

Indian Institute of Petroleum and Energy, Visakhapatnam

- Conducted statistical analysis to examine how global crises such as COVID-19 affected oil price volatility.

TOOLS, SKILLS & HOBBIES

Empirical Methods: Econometrics (OLS, IV, Panel, DiD)

Software/Language Proficiency: Python, R (basics), SQL, Stata, and LaTeX.

Tools/Databases/Libraries: Matplotlib, Seaborn, Tensorflow, Keras, scikit-learn, statsmodels, PyTorch, Selenium-Webdriver, BeautifulSoup, Flask, Matlab (basics), Oracle DB, Siebel CRM, Neo4J, GCP